



Coping knowledge boundaries between information system and business disciplines: An intellectual capital perspective



Jack Shih-Chieh Hsu^{a,1}, Tsai-Hsin Chu^{b,*}, Tung-Ching Lin^{a,1}, Chiao-Fang Lo^{a,1}

^a Department of Information Management, National Sun Yat-Sen University, Kaohsiung, Taiwan, ROC

^b Department of E-Learning Design and Management, National Chiayi University, Chiayi, Taiwan, ROC

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ABSTRACT

Information system development can be considered a collaboration between users and developers. The inability to leverage the localized knowledge embedded in these two stakeholders hinders software development work to achieve high performance. Exploring the ways to counter this difficulty is then critical. This study applies an intellectual capital perspective to address the issues around spanning the knowledge boundary between developers and users. Our findings highlighted how important effective knowledge boundary spanning is to both product and project quality. Furthermore, three dimensions of intellectual capital increased the degree to which knowledge boundary spanning was effective.

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1. Introduction

Multidisciplinary collaboration, such as cross-functional teams within an organization [11], or collaboration among organizations with different expertise [49], has become common in organizations for leveraging multiple knowledge bases to create innovations. Benefits are expected from multidisciplinary collaboration, including addressing the complexity of current phenomena [45], extending the solution space by bringing multiple perspectives [35], generating a wide variety of ideas and producing more creative designs [72,40]. However, challenges appear, especially when such collaborations cross the boundaries between specializations. For example, conflicts arise and collaboration may be dysfunctional when goal and value diversity are driven by professional differences [27], and communication can be ineffective as it always involves a long process of term definition negotiation in multidisciplinary collaboration [59]. These examples collectively highlight the not-to-be-ignored effect of the “knowledge boundary” in multidisciplinary collaboration [11,12]. *Knowledge boundary* refers to the knowledge delivery problems in which the tacit and

sticky nature of localized knowledge may actually hinder problem solving and knowledge creation across functions [57,9]. In practice, this specialization of knowledge increases the difficulty of collaborating across functional boundaries and accommodating knowledge developed in other practices [11].

In the information system (IS) area, a typical example of multidisciplinary collaboration is the IS development (ISD) process in which users and IS developers work together to counter requirements risks and generate better outcomes [38]. *Requirements risk* is the possibility that the elicited requirements will be of low quality, such as incorrect or invalid requirements [84,70]. Failing to elicit correct requirements in the design stage could increase the difficulty in the late stages of IS development [28,56]. Project teams may need extra resources and time to achieve predefined goals, and the developed system may not fully support users' daily work [64]. Studies have proposed ways to improve the quality of requirements elicitation, including development methodologies, tools and design paradigms [61,31]. In addition, empirical studies also emphasized the importance of including users in the requirements elicitation process to ensure success [36].

However, simply involving users in the system development process is far from sufficient. After decades of studying, academic researchers show inconsistent results on the effect of user participation [36,42]. Project performance may still be low even when users are included [47]. Even though contingency theory is widely adopted to explain the inconsistent findings [39], few studies attempt to answer this question from the process of or

* Corresponding author at: 85, Wunlong Village, Minsyong, Chiayi 62103, Taiwan, ROC. Tel.: +886 5 2263411x1511; fax: +886 5 2062328.

E-mail addresses: jackshsu@gmail.com (J.-C. Hsu), thchu@mail.ncyu.edu.tw (T.-H. Chu), tcclin@mis.nsysu.edu.tw (T.-C. Lin), aibatamoto@gmail.com (C.-F. Lo).

¹ Tel.: +886 7 5252000x4701; fax: +886 5 5254799.